Entry Test, Block Day 3

Saturday, 14 November 2026

Name _____ Points _____ / 20

20 multiple choice questions on days 1 and 2. Exactly one answer per question, tick the box. 15 minutes, no notes, no laptop.

Business Intelligence

6 points

1 The BI stack from day 1 has five layers. Which order is correct, from bottom to top?

- ☐ A Decisions, Reporting, Analysis, Warehousing, Raw data
- ☐ B Raw data, Warehousing, Analysis, Reporting, Decisions
- ☐ C Raw data, Reporting, Warehousing, Analysis, Decisions
- ☐ D Warehousing, Raw data, Analysis, Decisions, Reporting

2 Anna orders a coffee grinder, gets a voucher after a complaint, and buys again. Why did we call the BI stack a loop rather than a pipeline?

- ☐ A Every decision produces new raw data, which enters the stack again at the bottom
- ☐ B Reports are rebuilt every night
- ☐ C Dashboards refresh when the browser reloads
- ☐ D The warehouse deletes old data in a cycle

3 "Which product category is losing customers, and since when?" is a question for which layer?

- ☐ A Raw data
- ☐ B Data integration (ETL)
- ☐ C Data analysis and exploration
- ☐ D Decisions

4 Marketing keeps its own customer spreadsheet, support keeps another, and the numbers disagree. On day 1 we called this:

- ☐ A Operational BI
- ☐ B A data silo
- ☐ C A data-driven culture
- ☐ D Shadow BI done right

5 "How many orders will we get next week?" belongs to which kind of analytics?

- ☐ A Descriptive (what happened?)
- ☐ B Diagnostic (why did it happen?)
- ☐ C Predictive (what will happen?)
- ☐ D None, BI never looks forward

6 Of the four BI goals from day 1, which one is this: the emergency department plans staff per shift along the hourly patient curve?

- ☐ A Decide better
- ☐ B Earn more
- ☐ C Run cheaper
- ☐ D Know what is going on

Python and pandas

6 points

7 What is `[x * 2 for x in [1, 2, 3]]`?

- [ ] A `[1, 2, 3, 1, 2, 3]`
- [ ] B `[2, 4, 6]`
- [ ] C `6`
- [ ] D An error

8 For `prices = {"grinder": 89, "filter": 5}`, what is `prices["filter"]`?

- [ ] A `"filter"`
- [ ] B `5`
- [ ] C `1`
- [ ] D An error, dicts need numbers as keys

9 `df[df["amount"] > 100]` returns:

- [ ] A The number of rows with amount above 100
- [ ] B Only the rows whose amount is above 100
- [ ] C The column `amount` with values above 100 set to NaN
- [ ] D True or False

10 `df.groupby("country").agg(revenue=("amount", "sum"))` returns:

- [ ] A One row per order with the country added
- [ ] B One row per country with the summed amount in a column called `revenue`
- [ ] C The total revenue as a single number
- [ ] D An error, because `agg` needs a list

11 A column `order_date` with values like `2024/10/01` has dtype `object` (or `str`). What is true?

- [ ] A pandas already treats it as a date, you can group by month
- [ ] B It is text. You need `pd.to_datetime` before you can work with it as a date
- [ ] C `object` means the column contains numbers
- [ ] D The column must be deleted

12 `orders.merge(customers, on="customer_id", how="left")` keeps:

- [ ] A Only orders whose customer exists in `customers`
- [ ] B All orders, with customer columns filled where a match exists and NaN otherwise
- [ ] C All customers, even without orders
- [ ] D Only rows that exist in both tables

Database Architectures

8 points

13 CRUD stands for:

- [ ] A Create, Read, Update, Delete
- [ ] B Copy, Run, Undo, Deploy
- [ ] C Cache, Restore, Update, Drop
- [ ] D Create, Replace, Use, Dump

14 In a row-oriented database, finding all customers from Munich without an index is slow because:

- [ ] A The data is encrypted
- [ ] B The database has to scan every row and compare the city value
- [ ] C Row stores cannot filter by column
- [ ] D Munich is stored in a separate file

15 Adding an index on the `city` column:

- [ ] A Makes reads by city faster and writes slightly slower, because the index must be maintained
- [ ] B Makes both reads and writes faster
- [ ] C Has no cost at all
- [ ] D Replaces the primary key

16 A bank transfer moves 100 EUR from account A to account B. Which ACID property guarantees that either both bookings happen or neither?

- [ ] A Consistency
- [ ] B Isolation
- [ ] C Atomicity
- [ ] D Durability

17 A primary key:

- [ ] A Identifies each row uniquely and must not be empty
- [ ] B May repeat as long as the table is small
- [ ] C Is always a person's name
- [ ] D Is only needed in NoSQL databases

18 In an `orders` table, `customer_id` refers to the `customers` table. `customer_id` is:

- [ ] A The primary key of `orders`
- [ ] B A foreign key in `orders`
- [ ] C An index by definition
- [ ] D A transaction

19 Which statement about document databases (for example MongoDB) is correct?

- [ ] A They store each record as a flexible document, such as JSON, and do not require a fixed table schema
- [ ] B They require all records to have exactly the same columns
- [ ] C They cannot store nested data
- [ ] D They are the same as a file system

20 "No free lunch" in database design means:

- [ ] A Storage is never free
- [ ] B Every design choice trades something off, for example faster reads for slower writes
- [ ] C Backups are optional
- [ ] D Cloud databases are always cheaper